Removal of types

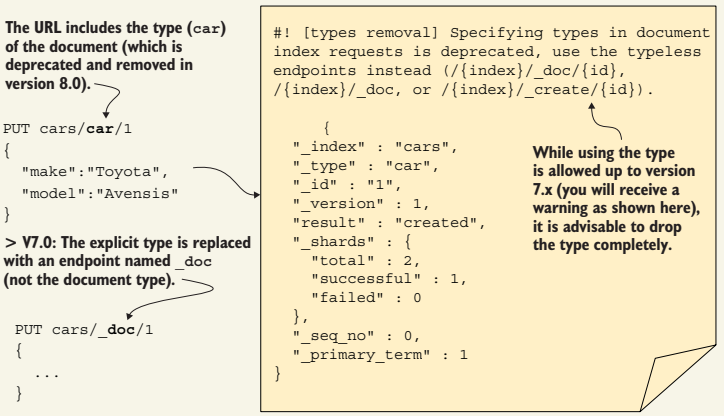
The data we persist has a specific shape: a movie document has properties related to a movie, data for a car has properties related to cars, and an employee document has data relevant to the context of employment and business. We index these JSON documents into respective buckets or collections: movie documents consisting of movie data need to be held in an index named `movies`, for example. Thus we index a document of type `Movie` into a `movies` index, `Car` into a `cars` index, and so on. That is, a movie document that we index into Elasticsearch has a type called `Movie`. Likewise, all car documents fall under the `Car` type, employee documents under the `Employee` type, and so on.

Prior to version 5.*x*, Elasticsearch allowed users to index multiple types of documents in a single index. That is, a `car` index could include types like `Cars`, `Peformance-Cars`, `CarItems`, `CarSales`, `DealerShowRooms`, `UsedCars`, or even `Customers`, `Orders`, and so on. While this sounds like a good plan to hold all car-related models in one place, there is a limitation.

In a database, the columns in a table are independent of each other. Unfortunately, that's not the case for Elasticsearch. Document fields may be various types, but they exist in the same index. That means a field in one type having a `text` data type cannot have a different type, such as `date`, in another index. This is because of the way Lucene maintains field types in an index. Because Lucene manages fields on an index level, there is no flexibility to declare two fields of different data types in the same index.

A single type per index was introduced beginning with version 6.0, so, for example, the `cars` index is expected to hold just `car` documents. When indexing a `car` document, the index name should be followed by the type: for example, `PUT cars/car/1` indexes a `car` document with ID `1` and puts it into a `cars` index.

However, the APIs were upgraded beginning with version 7.0.0: the document type is replaced by the default document type `_doc`, which is now a permanent fixture in the URL as an endpoint. Hence, the URL is `PUT cars/_doc/1`. As the figure shows, Elasticsearch allows us to use the types, but it throws a warning (as shown on the right in the figure) if our version is pre-8.x, advising us to use typeless endpoints.



Warning using pre-version-8 typed document indexing